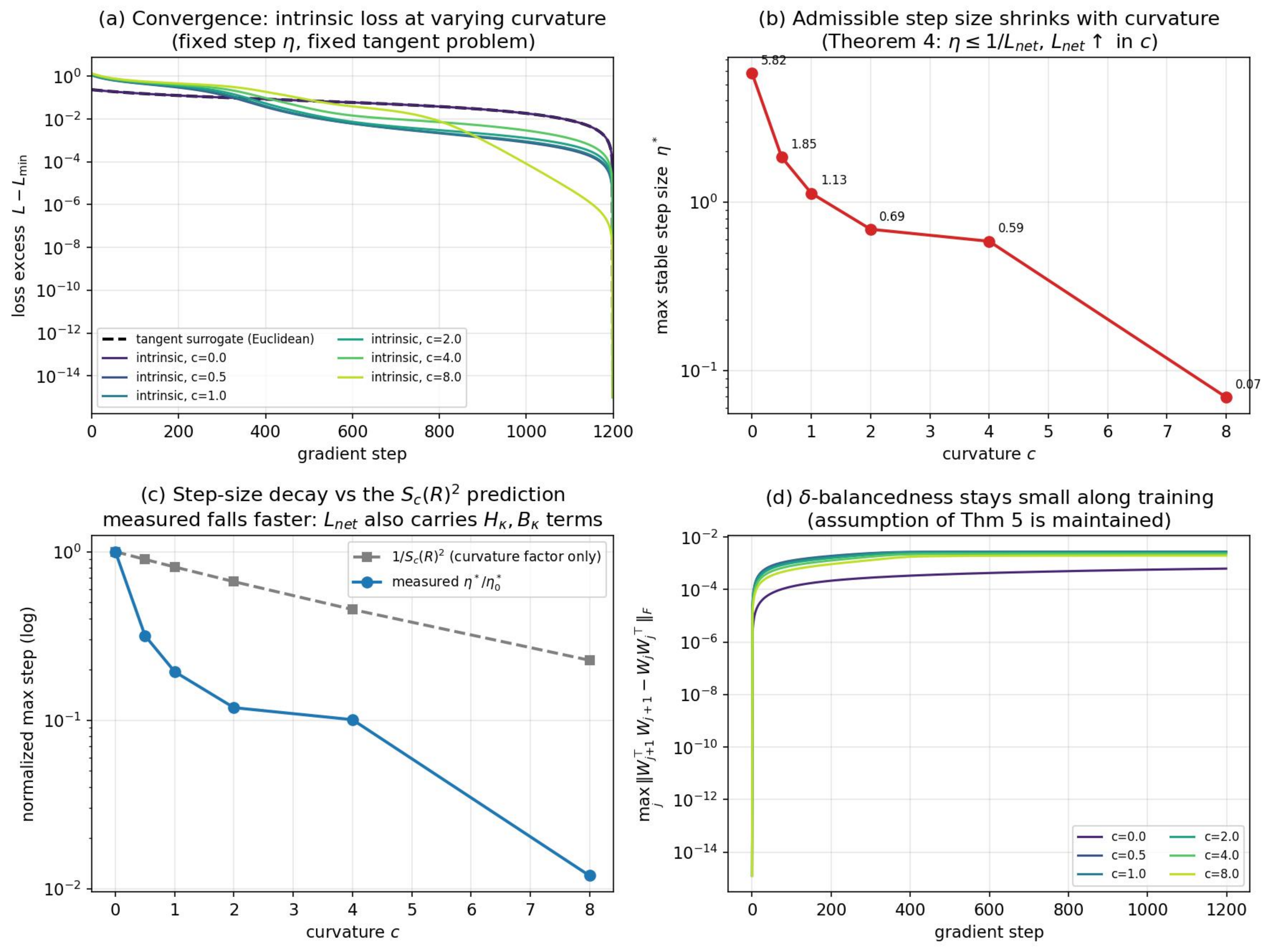

(a) Convergence: intrinsic loss at varying curvature
(fixed step η, fixed tangent problem)
loss excess $L - L_{\min}$
gradient step
tangent surrogate (Euclidean)
intrinsic, c=0.0
intrinsic, c=0.5
intrinsic, c=1.0
intrinsic, c=2.0
intrinsic, c=4.0
intrinsic, c=8.0
(b) Admissible step size shrinks with curvature
(Theorem 4: $\eta \leq 1/L_{net}$, $L_{net} \uparrow$ in c)
max stable step size η^*
curvature c
5.82
1.85
1.13
0.69
0.59
0.07
(c) Step-size decay vs the $S_c(R)^2$ prediction
measured falls faster: L_{net} also carries H_κ, B_κ terms
normalized max step (log)
curvature c
$1/S_c(R)^2$ (curvature factor only)
measured η^*/η_0^*
(d) δ-balancedness stays small along training
(assumption of Thm 5 is maintained)
$\max_j \|W_{j+1}^\top W_{j+1} - W_j W_j^\top\|_F$
gradient step
c=0.0
c=0.5
c=1.0
c=2.0
c=4.0
c=8.0